used for each type. As an example, in the United Kingdom plutonium containers made of wood and cadmium are limited to carrying 10 kg Pu; the container weighs 175 kg, is 1.3 m high, and 0.8 m in diameter.

21.4.2. Interim storage facilities

In the once-through fuel cycle, or with a limited capacity for reprocessing, separate spent fuel element interim storage facilities become compulsory. These facilities are either of the wet pool type (Czech Republic, Finland, Germany, Japan, Sweden, United Kingdom) or of the dry vault type (France, Germany, United Kingdom). The pool types consist of large water filled basins containing a geometrically safe (with regard to criticality) lattice of stainless steel racks for the spent fuel elements. The water is circulated for cooling. In general such pools are either located in a building at ground level or underground at 30–50 m depth. To increase pool capacity, boron steel plates are sometimes inserted thereby permitting denser packing of fuel elements without any criticality problem.

Zircaloy clad oxide fuel elements can be stored for decades in storage pools with very little risk of leakage. Metal fuels, especially those canned in magnesium or aluminum alloys, are less resistant and should not be stored as such in this manner for a prolonged time. The corrosion resistance of aluminum or magnesium clad fuel can be improved by electrolytic treatment yielding a protective oxide layer.

Some of the stringent requirements on the storage pools are: $k_{\text{eff}} < 0.95$ (even if unused fuel elements are introduced), earthquake safety, no possible water loss, water level automatically kept constant, adequate leakage and radiation monitoring systems, water temperature $<65^{\circ}$C, acceptably low radiation level in working areas, etc.

Dry vault storage may also be used. The fuel elements are stacked horizontally or vertically in concrete pipes which allows cooling of the fuel elements by air convection (forced or natural).

For long term storage ($\geq$10 y) the fuel elements must usually be recanned. For this purpose single fuel elements or bundles are placed in cylindrical containers, and the void is filled with some suitable material like lead, which has good heat conductivity and also provides some radiation protection. Depending on the external condition at the final storing place (humidity, temperature, etc.) the canisters are surrounded by an additional container to improve the lifetime of the fuel elements, which – preferably – should not be exposed to the biosphere until all radioactivity has disappeared.

21.5. ALTERNATIVE FUEL CYCLES

Fission energy can be obtained from uranium, using the *uranium once-through* option and the *uranium-plutonium fuel cycle*, and from thorium, by the *thorium-uranium fuel cycle*. Each fuel cycle offers a number of alternative routes with respect to reactor type, reprocessing,

and waste handling. Although the uranium based cycles are described with special reference to light water reactors, the cycles also apply to the old uranium fuelled gas cooled reactors.

In recent years several special fuel cycles have also been discussed which aim at the destruction of the minor actinides Np, Am and Cm in fast reactors. However, so far none of these have been implemented.

21.5.1. The uranium once-through (UOT) option

The heavy arrows in Figure 21.1 indicate the steps in the nuclear fuel cycle presently used on a large commercial scale. The cycle stops at the spent fuel interim storage facility; from here two alternative routes are available, one leading to the uranium-plutonium fuel cycle (reprocessing of the spent fuel elements, as described in the next section) and another leading to final storage of the unreprocessed spent fuel elements. The latter is referred to as the *once-through fuel cycle* (UOT) option.

In the UOT option, the energy content of unused ^{235}U, fertile ^{238}U, and fissile and fertile plutonium is not retrieved for future use, and the "waste" contains large amounts of these and other α-emitting nuclides. At present this is the cheapest option and also withholds plutonium from possible diversion to weapons use.

So far only four countries (Canada, Finland, Sweden, USA) have decided to use the UOT strategy, while all other nuclear power countries either are undecided or have elected to use the reprocessing strategy. The technical aspects of the UOT option are further discussed in §21.13.

21.5.2. The uranium-plutonium (U-Pu) fuel cycle

"Cycle" infers to some mode of recirculation of material. The term "fuel cycle" was originally used for the steps in which fissile and fertile material was isolated from used fuel elements (*reprocessing*) and returned to the front end of the process for use in new fuel elements, see Figure 21.1. The proper time to start reprocessing is a balance between loss of fissile material (^{241}Pu $t_{½}$ 14.4 y), economic interest loss on unused fissile and fertile material, and storage costs of the unprocessed fuels, on one hand, and, on the other hand, savings due to simplified reprocessing and waste handling. Originally a 180 d cooling time was considered appropriate (this time has been said to have been used in military programs). Presently the average cooling time for commercial fuel elements is 7–10 y because of lack of reprocessing capacity.

The fissile fraction in spent LWR fuel elements amounts to ~0.9% ^{235}U and 0.5–0.7% $^{239+241}Pu$. By recovering these and returning them to the LWR fuel cycle the demand for new uranium and enrichment services is reduced by ~30%. The uranium recovered may either be re-enriched and used in normal uranium oxide fuel or blended with the plutonium recovered to form *mixed oxide* (MOX) fuel elements (§21.1). MOX

fuel can also be made from recovered plutonium and depleted uranium. MOX fuel elements for LWRs contain up to 5% ^{239}Pu+^{241}Pu. Many tons of plutonium has already been used as MOX fuel in LWRs.

The re-enrichment of recovered uranium leads to a small contamination of the enrichment plant by ^{232}U, ^{233}U and ^{236}U, which are introduced as part of recovered uranium. Today, most batches of enriched uranium contain small amounts of these uranium isotopes. This phenomenon is usually called "rainbowing".

In this cycle "old" plutonium from earlier production is successively exposed to increasing neutron fluency, which changes its isotopic composition. This is shown in Table 21.6 for both virgin plutonium and virgin + recycled plutonium. The fissile fraction (^{239}Pu+^{241}Pu) decreases, requiring successively increased plutonium fractions in the MOX elements. The concentration of the most toxic isotope, ^{238}Pu, increases substantially. Its short half-life (88 y) results in a high specific radioactivity which increases the heat evolution and radiolysis of the reprocessing solutions. ^{238}Pu, ^{240}Pu, and ^{242}Pu all decay partly through spontaneous fission, with the emission of neutrons ($\sim 2 \times 10^6$ n kg^{-1} s^{-1}). This makes the MOX elements more difficult to handle. (All these changes also make the plutonium less suitable for weapons use.) ^{241}Pu has a critical mass of about half that of ^{239}Pu, so as ^{241}Pu builds up, criticality risks increase. The buildup of other actinides by recycling high mass isotopes in LWR fuel further increases reprocessing, handling, fuel manufacturing and waste problems.

Another possibility is to recycle only uranium, while the plutonium is left with the waste. This would produce a waste with a very high concentration of plutonium, except for highly enriched uranium fuel; the weight ratio of plutonium to fission products would be 10:35 (kg/t U, Table 21.2).

Table 21.6 The isotopic composition of plutonium (weight %) as function of number of recycles as MOX fuel in a PWR with a burnup of 33 000 MW_{th}d/t IHM per cycle

Recycles	^{236}Pu	^{238}Pu	^{239}Pu	^{240}Pu	^{241}Pu	^{242}Pu	^{244}Pu	Fraction left
0	7.2×10^{-7}	1.5	56.6	26.0	10.8	5.2	0.0004	1.00
1	4.6×10^{-8}	4.7	32.2	33.7	10.0	19.5	0.0013	0.47
2	6.6×10^{-8}	5.9	22.7	25.5	7.9	38.1	0.010	0.23
3	9.2×10^{-8}	4.4	19.8	20.2	5.0	50.6	0.020	0.12
4	1.1×10^{-7}	2.9	18.2	19.9	3.9	55.1	0.040	0.066
5	1.2×10^{-7}	2.3	17.6	20.3	3.6	56.1	0.080	0.036
6	1.2×10^{-7}	2.1	17.4	20.4	3.6	56.3	0.16	0.020
7	1.2×10^{-7}	2.1	17.4	20.5	3.5	56.3	0.29	0.011
8	1.2×10^{-7}	2.1	17.4	20.4	3.5	56.1	0.53	0.006
9	1.2×10^{-7}	2.0	17.3	20.4	3.5	55.8	0.96	0.003

The recycling is assumed to be in the form of 5% Pu and 95% depleted U.

The reuse of plutonium in LWRs would only be temporary if fast breeder reactors would become common (Fig. 21.1). FBRs are designed with a core, containing ~15% fissile Pu and ~85% ^{238}U (as depleted uranium) in the form of mixed oxides or carbides surrounded by a blanket of depleted uranium. In such a fast reactor, the presence of ^{235}U is undesirable as it reduces the neutron energy considerably by inelastic scattering. The actinide and fission product contents in discharged FBR fuel (core and blanket) and discharged LWR fuel are roughly the same on a GW_ey basis. Also the masses of total discharged fuel per GW_ey are comparable because the high burnup in the core is balanced by a very low burnup in the blanket. Breeding occurs only in the blanket (cf. §20.3). The burnup in the core elements is ~3 times higher than in LWRs, and the fraction of fission products is also 3 times larger. Since only a small part of the plutonium is burnt and the remainder has a high content of the fissile Pu isotopes, the used core fuel elements retain a high economic value, making it desirable to reprocess them after a short cooling time. Used FBR core elements may have a tenfold greater specific radioactivity than spent LWR fuel elements at the time of reprocessing and a much higher content of plutonium. Hence, criticality risks require a special reprocessing plant for core elements. The FBR blanket elements are simpler to handle because of a lower content of fission products and plutonium and they may be reprocessed in plants for LWR fuel elements. However, if the core elements are sufficiently diluted with blanket elements such a mix may be reprocessed in a "conventional" plant for LWR fuels.

In the FBR, ^{238}U is consumed both by fission (i.e. energy production) and by ^{239}Pu formation. Because the α-value of ^{239}Pu is 0.42, at least 70% ($100(1+\alpha)^{-1}$) of all ^{238}U is useful for energy production in the U–Pu cycle. This value should be compared with the fairly small fraction, $\leq$0.7%, of the natural uranium which is used in the UOT cycle (taking enrichment also into account), or $\leq$1% in the LWR MOX fuel recycle. The FBRs not only increase the useful energy of natural uranium by a factor of ~100, they also make it possible to burn current stockpiles of depleted uranium and also make it economic to mine low grade uranium ore, vastly extending the available uranium resources.

Currently, some 13 nuclear energy countries have decided to use reprocessing as part of their strategy for spent fuel management (Argentina, Belgium, Brazil, Bulgaria, China, France, Hungary, India, Japan, the Netherlands, Switzerland, Russia and the United Kingdom).

21.5.3. The thorium-uranium (Th-U) fuel cycle

Nuclear energy cannot be produced by a self-sustained chain reaction in thorium alone because natural thorium contains no fissile isotopes. Hence the thorium–uranium cycle must be started by using enriched uranium, by irradiation of thorium in a uranium- or plutonium-fuelled reactor or by using a strong external neutron source, e.g. an accelerator driven spallation source.

Fertile ^{232}Th can be transformed into fissile ^{233}U in any thermal reactor. The reactions in ^{232}Th irradiated by neutrons are given in Fig. 20.3. Of the thermally fissile atoms ^{233}U has the highest $\sigma_f/\sigma_{n,\gamma}$ ratio, i.e. highest fission efficiency. The η-value is high enough to permit breeding in the thermal region. Capture of neutrons in ^{233}U is not a serious drawback as a second capture (in ^{234}U) yields fissile ^{235}U, but reduces the breeding gain because two neutrons are consumed without a net increase in fissile material. Since σ_{tot} increases with neutron energy (from ~7 b at 0.025 eV to ~26 b for reactor conditions; cf. also Fig. 19.3) the slightly harder neutron spectrum in PHWR and GCR make such reactors the prime candidates for a thorium-uranium fuel cycle together with the molten salt reactor (§20.3). The fuel may be arranged in a core (^{233}U) and blanket (^{232}Th) fashion, or mixed fissile and fertile material as, for example, in the HTGR prototype (High Temperature Gas-cooled Reactor) graphite matrix fuels, or as a metal fluoride melt. The initial ^{233}U must be produced from thorium in reactors fuelled with ^{235}U (or ^{239}Pu) or in special accelerator driven devices. After sufficient amounts of ^{233}U have been produced, the Th-U fuel cycle may become self sustaining, i.e. thermal breeding is established.

The advantage of the Th—U fuel cycle is that it increases nuclear energy resources considerably because thorium is about three times more abundant on earth than uranium and almost as widely distributed. In combination with the uranium fuel cycle it could more than double the lifetime of the uranium resources by running the reactors at a high conversion rate (~1.0) and recycling the fuel. Very rich thorium minerals are more common than rich uranium minerals. The presence of extensive thorium ores has motivated some countries (e.g. India) to develop the Th—U fuel cycle.

No full-scale Th-U fuel cycle has yet been demonstrated and reprocessing has only been demonstrated on an experimental scale. The fuel cycle has to overcome the high activity problems due to the presence of ^{228}Th formed in the thorium fraction and ^{232}U formed in the uranium fraction (Fig. 20.3). The Th—U fuel cycle has a rather specific advantage over the U—Pu cycle in that its high active waste from reprocessing contains a much smaller amount of long-lived heavy actinides, although it probably contain some ^{231}Pa. With regard to nuclear weapons proliferation ^{233}U is almost as good a weapons material as ^{239}Pu and easier to produce as a single isotope by continuous withdrawal of protactinium, since it is the decay product of ^{233}Pa ($t_{½}$ 27 d), see also §21.7.

21.6. REPROCESSING OF URANIUM AND MIXED OXIDE FUELS

The main purpose of commercial reprocessing is:

(1) to increase the available energy from fissile and fertile atoms;

(2) to reduce hazards and costs for handling the high level wastes.

Two other reasons are sometimes mentioned:

(3) to reduce the cost of the thermal reactor fuel cycle;

(4) to extract valuable byproducts from the high active waste.